Jiayu Shi

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Address: 564 Centennial Olympic Park Dr NW, Atlanta, GA, 30313

OBJECTIVE

Electrical Engineering major specializing in circuit technology and robotics. Adept at designing analog and digital circuits, writing ARM and RISC-V assembly, and leveraging proven executive leadership experience to guide cross-functional teams toward executing complex technical projects. Experienced in building custom dual-extruder 3D printers and conducting advanced polymer materials research. Seeking a May 2027 hardware engineering internship.

EDUCATION

Georgia Institute of Technology | Atlanta, GA *Expected Graduation: May 2028*
Bachelor of Science in Electrical Engineering | GPA: N/A (First Semester)
Transfer College: Sichuan University | Chengdu, Sichuan, China *Sep 2024 – Jul 2026*
Bachelor of Engineering in Electrical and Computer Engineering | GPA: 3.82/4.0

SKILLS

- Programming: C, C++, MATLAB, ARM Cortex-M4 Assembly
- Hardware: STM32 Microcontrollers, Stepper Motors, Matrix Keypads, Oscilloscopes
- Circuit Design: Analog/Digital Circuits, Op-Amps, Transistor Biasing (MOSFET/BJT), RC/RLC Filters
- Software/Tools: PSpice, Keil µVision 5, Pronterface, G-code
- Languages: Mandarin (Native), English (Fluent)

EXPERIENCE

Shenzhen Bestsoon Electronic Technology Co., Ltd. | Shenzhen, China *Jul 2026-Aug2026*
Intern / Technical Department

- Mastered the basic structure, working principles, basic operation, and maintenance procedures of the ASMPT AD8312PIUS Die Bonder (chip mounting machine).
- Applied academic coursework to practical tasks within the technical department, collaborating with colleagues to consistently complete technical assignments with high quality.

PROJECTS

Research on Polymer Light-Responsive Materials Based on 3D Printing *Mar 2025 – Present*
Role: Project Leader / Supervisor: Wenwen Xu

- Engineered a custom dual-extruder Direct Ink Writing 3D printer using BigTreeTech Octopus Pro and TMC drivers, optimizing firmware and G-code to maximize print precision.
- Developed custom dual-extruder slicer software enabling concentration gradient and multi-material zoned printing, integrating Cura 3D and Pronterface to centralize the slicing, previewing, debugging, and printing workflow.
- Formulated photo-responsive inks (azobenzene, PVP, silica) to meet strict printability and light-actuation metrics.
- Directed team research on latch-based elastic energy structures, achieving macroscopic snapping transitions.

STM32 Embedded Systems Development *Mar 2026 – Jul 2026*
Role:Project Leader / Supervisor: Fashu Xu

- Programmed STM32L476xx microcontrollers in C and ARM assembly, implementing register-level peripheral configurations, real-time interrupts, and I2C hardware interfacing.

RELEVANT COURSEWORK

Digital Circuits & Systems, Linear Circuits & Systems, Embedded Processors & Interfacing, Microelectronic Circuits

LEADERSHIP

Student Council, Sichuan University–Pittsburgh Institute (SCUPI) *Jul 2025 – Jul 2026*
President

- Directed a 55-member executive team to plan and execute campus-wide academic and cultural initiatives, prominently featuring the Student Innovation Forum and Academic Integrity Campaign.
- Strategically managed a $12,000 annual budget across 5 major campus events, directing the procurement and rental of essential event infrastructure, including stage lighting, sound systems, and custom props.
- Spearheaded large-scale events—peaking at over 600 attendees for the Glowing Fair New Year Market—by coordinating cross-departmental collaboration, establishing timelines, and resolving on-site contingencies.

"Protecting the Plateau Sapphire" Summer Social Practice Program *Jul 2025 – Aug 2025*
Team Leader

- Led a 10-person research team in a mixed-methods field initiative on Qinghai Lake's ecological governance, collecting and analyzing 500 survey samples alongside official provincial datasets.
- Authored a bilingual ecology handbook based on field findings that secured Outstanding Team, Individual, and Practice Report awards.